
Detecting Failures in LLM Agents: What Lightweight Runtime Verification Can and Cannot See

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Verifying an agent after it answers is too late, and verifying every step with a
2 second model costs more than the agent. We ask what is verifiable from *observable*
3 *step telemetry alone* — semantic embeddings of step output, token uncertainty,
4 action metadata — by monitors that cost microseconds and are fitted on healthy
5 runs only, and where that cheap signal provably runs out. On 2,823 committed
6 episodes across three frameworks, four agent models and a commercial API, a
7 one-class echo-state-network CUSUM ensemble detects 0.71 ± 0.07 of failures at
8 a 5% false-alarm budget. Two findings bound it. First, a *horizon law*: the temporal
9 monitor’s advantage over a memoryless baseline grows from +0.09 detection
10 points at ≤ 3 post-onset steps to +0.40 at ≥ 9 , and it predicts its own failure region
11 on an external corpus we did not build: on AFTraj-2K, where 94% of failures end
12 within eight steps of onset, pooled detection is 0.048 exactly as the law forecasts,
13 while at ≥ 9 steps it is 0.509. Second, an *observability boundary*: a wrong-but-
14 well-formed value perturbs no behavioural channel, so no reference-free monitor
15 can see it. We therefore pair the monitor with deterministic checks that need no null
16 and no threshold — recomputing a run’s stated total from the tool results it actually
17 received — which catch 60% of failures at **0 of 63** false positives against the
18 monitor’s 54% at 17%, transfer across model families unchanged (110/110 at 0/10),
19 and close into repair that lifts task success from 52% to 73%. The contribution is a
20 measured decomposition of which cheap observable verifies which failure, and an
21 explicit statement of what neither verifies.

22 1 Introduction

23 An agent episode is a sequence of steps, each emitting observable telemetry: what the agent said, how
24 confident its tokens were, which tool it called, what came back, how long it took. The verification
25 question is whether that stream is enough to decide, *mid-run*, that something has gone wrong — early
26 enough to halt, roll back, or repair.

27 Two answers dominate. Check the final output, which is cheap but arrives after the budget is spent; or
28 judge every step with a second LLM, which is accurate and costs a model call per step. This paper
29 measures a third position: a one-class monitor over step telemetry at $\sim 200 \mu\text{s}$ per step, fitted on
30 healthy runs only, with no failure labels and no second model. One optional variant — the logistic
31 fusion of §5 — is supervised and needs on the order of twenty labelled failures; every other monitor
32 and every check reported here is label-free. We are interested less in headline accuracy than in the
33 *boundary* — what such a verifier structurally cannot see, and what to pair with it there.

34 Contributions.

1. **A horizon law with external validation.** The temporal monitor’s advantage over a memoryless baseline is governed by post-onset runway, not by detector identity. The law predicts where the monitor will fail, and holds on a corpus built by another group (§6). Only one of our two external corpora can test it: the other labels trajectories rather than steps, so it has no onset and no horizon.
2. **A measured observability boundary.** Content corruption that changes data without changing behaviour is invisible to behavioural telemetry; we show richer embeddings do not fix it, and that a deterministic grounding channel does (§8).
3. **Deterministic verification that needs no null.** Checks that recompute a run’s answer from the tool results it received beat the monitor on precision by construction — 0% false positives against 17% — and transfer across model families with nothing retuned (§9).
4. **Verification closed into repair,** with the negative results kept: rolling a flagged run back and re-running it lifts task success from 52% to 73%, but only one of five failure classes is actually repairable (§9).
5. **A measured LLM judge,** not a stipulated one. The judge everyone escalates to scores $p_{\text{detect}} = 0.548$ where analyses assume 0.90; substituting the measured value drops our own escalation claim from 83% to 44% recovery, and we report the lower number (§7).

2 Telemetry and the Monitoring Problem

An episode is a sequence $\mathcal{E} = \{(q_t, a_t, o_t, r_t)\}_{t=1}^T$ of reasoning q_t , action a_t , observation o_t and runtime metadata r_t . At each step the adapter emits a causal feature vector

$$\mathbf{x}_t = [\mathbf{e}_t; \mathbf{u}_t; \mathbf{m}_t; \mathbf{g}_t] \in \mathbb{R}^d, \quad (1)$$

using only information available at or before t : $\mathbf{e}_t$ a 32-dim deterministic character-trigram hash of the step text (no model, no network), $\mathbf{u}_t$ four token-surprisal aggregates where the provider exposes logprobs, $\mathbf{m}_t$ action type, latency, output length, retry and error metadata, and $\mathbf{g}_t$ nine content-grounding features (§8). Monitors are fitted on healthy episodes only and standardised with healthy statistics, $\tilde{\mathbf{x}}_t = \mathbf{D}^{-1}(\mathbf{x}_t - \boldsymbol{\mu})$.

For a failed episode with onset τ , a detector alarms at $\hat{\tau} = \inf\{t : s_t > \gamma\}$. We report detection at a *realized* false-alarm rate rather than AUROC alone: a monitor that ranks well can still alarm after the run ends or interrupt too many healthy runs, and both matter for a verifier that is allowed to halt.

3 Streams and the Serving Decision

No single detector observes every failure mechanism, so the system is a set of independently calibrated streams.

Temporal stream. An ensemble of $K=8$ leaky echo-state reservoirs [3] predicts the next standardised step; member k has state $\mathbf{h}_t^{(k)}$ and surprise $e_t^{(k)} = \|\tilde{\mathbf{x}}_t - \hat{\mathbf{x}}_t^{(k)}\|_2$. The ensemble exposes mean surprise and disagreement, which are robustly z -scored and accumulated by a one-sided CUSUM [6],

$$C_t = \max\{0, C_{t-1} + z_t - \kappa\}, \quad (2)$$

so that weak but persistent deviations integrate rather than being smoothed away. Everything in (2) depends only on $\mathbf{x}_{0:t}$ and fit-time quantities: strictly causal, no lookahead.

Abrupt-change stream. Accumulation is useless when one or two steps remain after onset. A memoryless Δ -Mahalanobis stream scores healthy step differences $\Delta\mathbf{x}_t$ against a shrinkage covariance [4], and covers exactly the regime the temporal stream cannot (§5).

The wrapper, not the reservoir, carries the margin. The temporal stream is presented as an ESN because that is what we ship, but the credit belongs elsewhere and the measurement says so. Giving a GRU the same per-channel max-fusion wrapper lifts it to detection **0.76** at AUROC 0.873, past the ESN’s 0.71 / 0.872; a linear vector-autoregression leads the ESN by roughly 1.9 steps of lead while trailing badly on detection. The transferable contribution is the wrapper — score each channel

separately and take the max, rather than scoring a pooled vector. The reservoir stays the default for two practical properties rather than for accuracy: a lower healthy false-alarm rate (0.069 against 0.113) and a fit roughly $100\times$ faster, which is what makes the per-deployment recalibration of §4 affordable at all.

Calibration and the serving decision. Each stream’s threshold is a quantile of per-episode maxima over *held-out* healthy episodes,

$$\gamma_j = Q_{1-\alpha_j} \left(\left\{ \max_t s_{j,t}^{(i)} : \mathcal{E}_i \in \mathcal{H}_{\text{val}} \right\} \right), \quad (3)$$

and the served alarm is a per-stream OR rather than a fused score,

$$A_t = \bigvee_j \mathbb{I}[s_{j,t} > \gamma_j] \vee B_t^{\text{contract}} \vee B_T^{\text{complete}}. \quad (4)$$

Equation (4) is a design result, not a formality. Under a single shared threshold only episode-max ordering matters, so the stream with the heavier healthy tail sets the bar for both: measured, one behavioural healthy episode destroyed **16 of 17** grounding-only detections of context corruption. No monotone rescaling fixes an ordering problem; independent thresholds do: no stream’s evidence can be numerically cancelled by another’s, which held at every seed we ran. The cost is paid in false alarms — an OR over J streams admits each stream’s budget — so this is a non-degradation property bought with alarm rate, not a free guarantee.

4 Real-Ecosystem Validation

Everything claimed about deployment rests on 2,823 committed episodes over 25 corpora — three frameworks, three local models across two families, and the gemini-2.5-flash API — each a live agent run against real tools, with telemetry measured at the adapter rather than constructed. Looping and tool cascade are caught almost everywhere (0.48–1.00) and goal drift wherever it is present (0.66–0.86 on four of five corpora); the weak spots are the content-corruption classes, `wrong_document` (0.00–0.20) and `malformed_json` (0.05–0.50), which is what motivates §8 rather than a rationalisation of it.

The breadth claim rests on the most filtered corpora. An episode is discarded when it is too short to score or when its injection never landed, and the rate is nowhere near uniform: `langgraph` **55.4%**, `real` **53.8%** and `ollama_llama8b` **49.2%**, against `real_research7b` at **1.7%**. The corpora carrying the cross-framework and cross-model claims are precisely the ones where most attempts were discarded, so those breadth results rest on a more selected sample than the headline numbers do. The filter is also asymmetric: of the four rejection rules, only the length rule can reject a healthy episode, and the other three fire solely on injected ones because they check that the injection actually landed — positives are filtered on a condition negatives are never tested against. Every rejection is written to its corpus’s `rejected.json` with a reason, and the per-corpus and per-rule tables are released with the data.

Two structural findings shape the rest. First, **monitors do not transfer across deployments**: calibrated on `qwen2.5:7b` and deployed on `llama3.1:8b` over a matched task, tool and injector plan, the primary monitor sits at chance (AUROC 0.527, healthy FA 0.75), while the *same* target recalibrated on itself reaches 0.885. That is a property of the calibration, not of a hard corpus, and it means a verifier of this kind is per-deployment by construction. Second, the memoryless baseline wins outright on short-episode datasets — which the next section explains.

5 The Horizon Law

Over 1,002 injected episodes the ESN–Mahalanobis detection gap is $+0.09$ at post-onset horizon ≤ 3 steps, $+0.14$ at 4–8 and $+0.40$ at ≥ 9 (correlation $+0.25$). The reservoir needs post-onset steps to integrate evidence: where it has them it wins decisively, and where it does not its margin collapses and the memoryless distance becomes competitive. This is a statement about *regime*, and it is what makes the monitor’s failures predictable rather than mysterious.

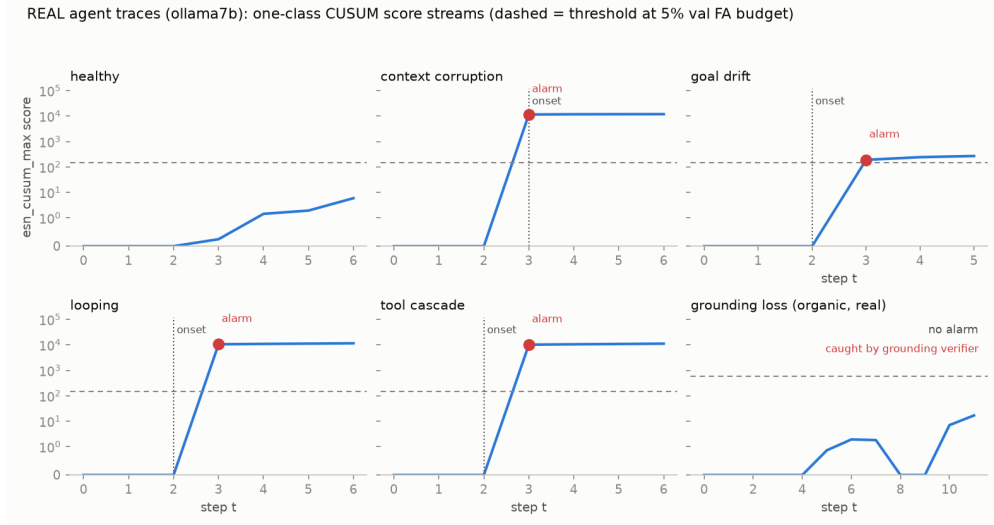


Figure 1: **What the verifier sees.** One-class CUSUM streams per failure class on real agent traces (qwen2.5:7b against real tools); dashed = threshold at the 5% validation budget, dotted = verified onset. Healthy stays two orders of magnitude below the line and every injected class alarms one step after onset. The sixth panel is different in kind: fabrication cannot be injected into a live run, so it is a genuine organic episode — the CUSUM stays flat, because a fabricated figure perturbs no behavioural channel, and the class is caught instead by the deterministic verifier of §9. That is the observability boundary shown rather than asserted. The y axis is symlog: the CUSUM accumulates multiplicatively once a failure takes hold.

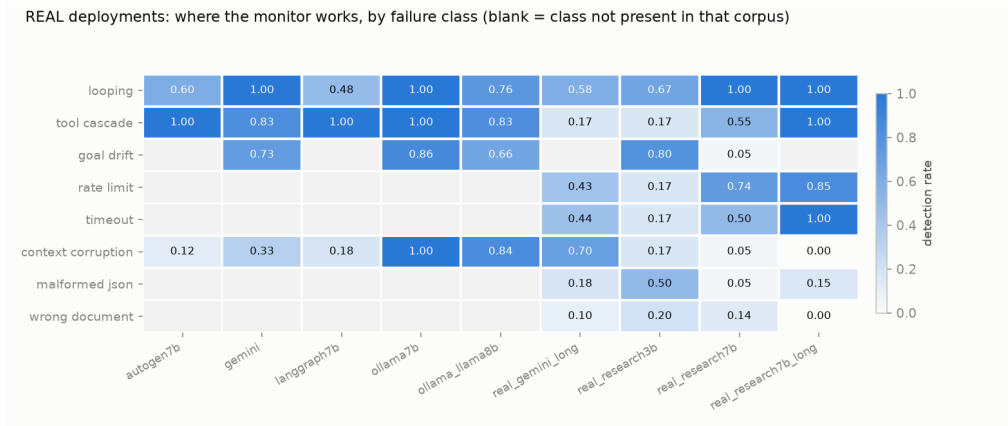


Figure 2: **Where the monitor works, by failure class**, across nine real corpora; blank where a corpus does not contain that class, because absent and undetected are different claims. Looping and tool cascade are caught almost everywhere and goal drift wherever present; the weak spots are exactly the content classes that motivate §8.

123 **The fusion buys robustness, not dominance.** A cross-fit logistic fusion of the two parents reaches
 124 grand-mean AUROC 0.826 against 0.802 (ESN) and 0.807 (Δ -Mahalanobis), because each parent
 125 collapses on some dataset and the fusion never does. Measured against whichever parent is better *on*
 126 *that dataset*, however, it is at or below it on 7 of 8 datasets in AUROC (mean -0.014) and on all 8 in
 127 detection rate (mean -0.140).

128 The robustness that buys is real but conditional, and §6 contains its counterexample: on ATBench
 129 the memoryless parent is at chance (0.457) and drags every 50/50 blend down with it (0.438–0.463)
 130 against the ESN alone at 0.779, while on AFraj the same fusion is marginally ahead ($+0.015$).
 131 Fusion protects against *a* parent being weaker; it does not protect against one being uninformative,

132 because a 50/50 blend has no way to notice. The hybrid is a reasonable default when the regime is
133 unknown *and* both parents are known to carry signal — not a general safety net, and not a dominance
134 result.

135 6 External Validation on Corpora We Did Not Build

136 Every number above comes from corpora we collected, so none of them says whether the benchmark
137 is hard. We therefore ran the monitors, unchanged and fitted on each corpus’s own healthy runs,
138 against two external benchmarks.

139 **AFTraj-2K** [11] contributes 1,882 multi-agent trajectories after the same $T \geq 4$ filter (1,111
140 healthy, 771 failed) with a judge-annotated earliest-decisive-error step. *Ranking transfers zero-shot*:
141 episode AUROC **0.745** on another group’s agents and frameworks, with the uncertainty channel
142 absent entirely, against 0.802 pooled on our own. *The horizon law predicted the operating point*,
143 *and held*: detection is **0.509** on failures with ≥ 9 steps of runway against 0.170 for the memoryless
144 baseline, but only 53 of 771 AFTraj failures have that much room and 61% end within three steps
145 of onset, so pooled detection is **0.048**. That pooled figure is the mechanism’s prediction, not an
146 excuse offered after seeing it: §5 fixed the relationship on our corpora, and AFTraj supplies a failure
147 distribution concentrated where the relationship says detection is impossible. A law that forecasts its
148 own failure region and then survives a corpus it never saw is stronger evidence than a higher pooled
149 number would have been.

150 **ATBench** [8] contributes 604 trajectories (279 safe, 325 unsafe) labelled under a different taxonomy
151 of risk source, failure mode and harm. Here the monitor *alarms*: AUROC **0.779** at detection **0.311**,
152 because ATBench’s unsafe behaviour is a pattern of action — an over-privileged call, an unvalidated
153 tool result — rather than a single wrong value. Two independent confirmations fall out. The fusion
154 collapses to chance (0.457 for the memoryless parent, dragging every 50/50 blend to 0.438–0.463
155 against the ESN’s 0.779 alone), which is §5’s pooled-not-dominant finding shown on foreign data.
156 And per failure mode the split is sharp — over-privileged actions 0.508 and unvalidated tool outputs
157 0.473 against inaccurate information **0.038** — reproducing, on labels we did not write, exactly the
158 content blind spot §8 exists for.

159 For scale: an LLM auditor on AFTraj reports 66.44 Exact-F1 using a model call at every step [11].
160 We do not match that and do not claim to; detection at a false-alarm budget is not Exact-F1. The
161 comparison we can make is cost: 173 μ s and no model call, against one 7B forward pass.

162 7 The Judge Is a Verifier Too, and It Should Be Measured

163 The default answer to “who verifies the agent” is another LLM. Escalation analyses — ours included
164 — usually *stipulate* that judge’s operating point. We measured it instead. A gemini-2.5-flash judge on a
165 labelled subset of 161 distinct prompts scores $p_{\text{detect}} = \mathbf{0.548}$ (95% CI 0.44–0.65) and $p_{\text{false}} = \mathbf{0.052}$,
166 against the 0.90/0.02 the analysis assumed; both stipulated values lie outside their measured intervals.
167 Substituting the measured rates and re-running all five master seeds with everything else fixed, the
168 *call saving survives* (6% of judge-every-step calls) but detection recovery falls from 83% to **44%**,
169 supported at **0 of 5 seeds**. We report the escalation layer’s detection claim as 44%.

170 The layer is still worth having, for a reason that only appears once both are measured on the same
171 episodes: judge and monitor fail on *different* classes (Figure 3). The judge is perfect on goal
172 drift (21/21) and nearly blind on context corruption (0.18), where the monitor is the reverse (1.00).
173 That is the empirical case for escalation as a complementary verifier rather than a more expensive
174 approximation of the cheap one — and a caution against citing a stipulated judge as a baseline.

175 8 The Observability Boundary and the Grounding Channel

176 Behavioural monitors share one blind spot: corruption that changes *data* without changing *behaviour*.
177 A wrong-but-well-formed number perturbs no channel in (1), so no reference-free detector can
178 reject it. The *g* channel adds nine causal content features — query–result dissimilarity, result self-
179 consistency, JSON-prefix validity, character anomaly, and a binary lexical-relevance flag costing 2 μ s

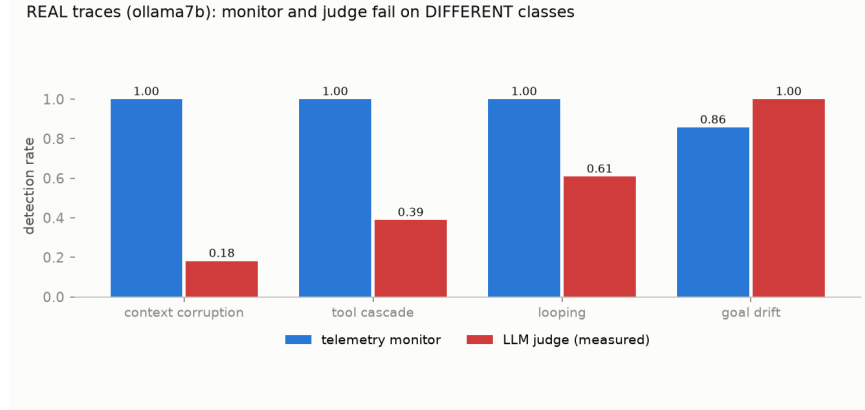


Figure 3: **Monitor and judge fail on different classes**, both measured on the same real corpus. The judge is perfect on goal drift and nearly blind on context corruption; the monitor is the reverse. Complementary verifiers, not substitutes.

180 — which makes query–evidence–reasoning *inconsistency* observable without claiming to observe
 181 truth. Pooled over 602 injected episodes, the content gate lifts content-class detection $0.23 \rightarrow \mathbf{0.40}$
 182 and behavioural detection $0.73 \rightarrow \mathbf{0.80}$, gaining 37 and 25 discordant detections against zero lost.

183 The boundary is real and we measured its floor: on a plausible-corruption set, richer MiniLM
 184 embeddings leave detection at **0.00** while behavioural AUROC falls 0.82 to 0.70, at 4.4 ms per text
 185 and 90 MB. The failure is observability, not feature capacity — which is the argument for §9 rather
 186 than for a bigger encoder.

187 9 Deterministic Verification, and Repair

188 The monitor needs a healthy null per deployment and spends part of a false-alarm budget. A
 189 check needs neither. `total_consistency` recomputes a run’s stated total from the tool results
 190 *that run actually received*; `required_coverage` confirms every call the task requires was made;
 191 `tool_contract` asks whether a result matched any shape its tool can return, and so reports when
 192 the result arrives rather than at the answer. None reads the hidden world the task was generated from.

Table 1: Checks against the behavioural monitor, same episodes and same objective labels. Recall is comparable; the difference is precision.

	checks	monitor
failures caught, $T=0.2$ (served)	60% (96% with coverage)	54%
false positives, $T=0.2$	0/63 = 0%	11/63 = 17%
failures caught, $T=0.9$	65% (96% with coverage)	40%
false positives, $T=0.9$	0/38 = 0%	6/38 = 16%

193 The checks were written by inspecting failures in the serving arm, so that arm cannot also be their
 194 test set. A further 120 episodes at disjoint task seeds were collected afterwards and scored with the
 195 checks frozen: 54% caught by totals, **93%** with coverage, and **0 of 64** false positives. A llama3.1:8b
 196 arm on the *same* task seeds, with nothing retuned, catches **110 of 110** at **0 of 10** — the checks
 197 transfer where the monitor does not, because they carry no calibration. Across every labelled corpus
 198 `tool_contract` trips on **0 of 1825** healthy episodes and flags 46% of context corruption, **215 of 218**
 199 within one step of onset.

200 **Repair.** Each flagged episode is rolled back to its last fact-gathering step and re-run, paired on
 201 the identical prefix and task; the rollback is real, and success is graded by an oracle the repair
 202 prompt never sees. Naming which check failed — without supplying any value — recovers **45%** of
 203 genuinely-wrong runs against a 16% resampling control ($p = 0.0005$), which nets task success from

204 **52% to 73%** with zero correct runs broken. Two negative results sit inside that: routing the step to a
 205 calculator the agent already holds does *not* beat retry luck ($p = 0.17$), and supplying the recomputed
 206 answer buys nothing — 26 of 55 value-bearing hints contain the correct total outright, while the
 207 value-free rung recovers at least as much. The recovery does not come from handing over the answer.

208 Coverage is partial and the shape of the gap is measured: over five injection classes $\times$ five seeds,
 209 every behavioural alarm is followed by a repair attempt (21 of 21), but `goal_drift` is the only class
 210 a retry fixes (4 of 5). Where the tool layer itself is broken a retry fetches the same broken result, and
 211 the value of the intervention is ending the episode fast — a loop trap exits at exactly 10 steps against
 212 30 without the circuit breaker.

213 10 Limitations

214 Stated as measured. (1) **No cross-deployment transfer**: the healthy null must be collected under the
 215 exact serving distribution; transferred across a model family the monitor sits at chance (0.527) while
 216 the same target recalibrated reaches 0.885. (2) **Slow goal drift** evades every per-step surprise monitor
 217 tested; the limit is the *rate* of change, not the class — abrupt goal changes are caught at 0.66–0.86.
 218 (3) **Content detection is bounded by telemetry completeness**, not by logprob access: ablating
 219 the token-surprisal channel moves the ESN by AUROC +0.000. (4) **Plausible-value corruption**
 220 is undetectable from telemetry by construction. (5) **Fabrication claims are limited by base rate**:
 221 9 of 175 organic episodes are labelled hallucinated and only 2 are fabricated *inputs*, below the
 222 pre-registered minimum of 10, so no detection claim is made in either direction; under provocation
 223 the grounding verifier catches 0.55 with no healthy false positives — specific, but half sensitive.
 224 (6) **Scope**: mock-tool and research-loop tasks, two local model families and one commercial API.

225 11 Related Work

226 **Concurrent online failure prediction**. Detecting failure during an episode is no longer an unoccupied
 227 position. AgentForesight reframes attribution as online auditing and releases AFTraj-2K [11]; weakly
 228 supervised early alerting learns turn-level risk from trajectory labels alone [1]. Both use an LLM
 229 auditor. We therefore do not claim mid-episode detection from observable telemetry as novel; what is
 230 ours is the cost class and the boundary analysis.

231 **Internals versus observables**. Hidden-state probes report predicting failure earlier than any
 232 observable-behaviour monitor [7]. We do not dispute it: with the weights, activations are richer. The
 233 premise here is the deployment where you do not hold them — a hosted API, a third-party agent —
 234 and there the choice is between telemetry and nothing.

235 **Runtime enforcement** intervenes during execution but requires a specification of what unsafe means:
 236 rule sets [9] or labelled unsafe states [10]. Ours is label-free and specification-free, which is why
 237 it transfers badly and needs no rule authoring. **Post-hoc attribution** asks which step caused a
 238 failure after the fact and finds it hard — 14.2% accuracy on the responsible step [12]; that problem
 239 is outcome-conditioned, ours is not. **Hallucination detection** needs model access or repeated
 240 sampling [5, 2]; tool-using agents admit the cheaper deterministic route of §9.

241 12 Conclusion

242 A cheap verifier over step telemetry covers more of the agent-failure space than its cost suggests, and
 243 its failures are predictable rather than arbitrary: the horizon law says where it will not fire, and held
 244 on a corpus built by another group. Where it structurally cannot see — a wrong-but-well-formed
 245 value — the answer is not a bigger encoder but a different mechanism, and a deterministic check that
 246 recomputes the answer from the evidence the run received is both more precise and more portable
 247 than the monitor it supplements. We think the useful framing for agent verification is coverage: which
 248 cheap observable verifies which failure, measured, with the gaps stated.

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A Appendix

A.1 Telemetry versions

Version	Dims	Content
v1	43	e_t 32-dim hash embedding; u_t 4 token-surprisal aggregates; m_t action one-hot, log latency, output length, error
v2	43	Tool results appended into step text so corrupted results reach the semantic channel
v3	51	8 derived dims: cosine drift, task-anchor similarity, tool success, retry, per-tool latency, context ratio, reasoning depth, self-consistency
v4	60	9 content-grounding dims: query–result dissimilarity, result self-consistency, JSON-prefix validity, character anomaly, lexical flag

Table 2: Causal telemetry evolution. v4 costs 608 μ s/step at the adapter; the lexical flag alone costs 2 μ s/result.

dataset	ESN	Δ -Maha	better parent	logistic	Δ
sim	0.890	0.786	0.890	0.889	-0.000
autogen7b	0.833	0.774	0.833	0.777	-0.056
langgraph7b	0.828	0.885	0.885	0.884	-0.002
ollama7b	0.994	0.895	0.994	0.982	-0.013
real_research3b	0.556	0.665	0.665	0.643	-0.022
real_research7b	0.777	0.848	0.848	0.847	-0.001
real_research7b_long	0.790	0.849	0.849	0.857	+0.008
gemini	0.749	0.756	0.756	0.731	-0.025
grand mean	0.802	0.807	0.840	0.826	-0.014

Table 3: Episode AUROC. The fusion beats either parent’s grand mean while sitting below the per-dataset better parent on 7 of 8.

corpus	monitor	AUROC	detection	FA
AFTraj-2K	esn_cusum_max	0.745	0.048	0.023
AFTraj-2K	hybrid_weighted50	0.760	0.047	0.018
ATBench	esn_cusum_max	0.779	0.311	0.071
ATBench	Δ -Mahalanobis	0.457	0.268	0.161
ATBench	hybrid_weighted50	0.463	0.277	0.179

Table 4: Monitors run unchanged on two external corpora, fitted on each corpus’s own healthy runs. Neither carries token logprobs, so both run on $e+m+x$ only.

283 A.2 Per-dataset hybrid results

284 A.3 External corpora

285 A.4 Reproducibility

286 Code, all 2,823 committed traces, every result table and the figures are publicly released under a
287 permissive licence; the URL is withheld for review. The release also carries a standalone contributions
288 document that states each scientific result in the project — the laws, the measured boundaries, the
289 verification and repair outcomes, the negative results — next to the artifact it is computed from and
290 the bound it carries, so a reader can audit the set rather than reconstruct it from the paper. Each
291 headline number in this paper is tied to a named artifact by a claim ledger that recomputes it from
292 the committed file on every run, and a SHA-256 manifest covers every source file, trace and result
293 so a reader can verify the number in the text came from the file in the repository. An end-to-end
294 behavioural snapshot re-runs the study at a fixed seed and compares every value against a stored
295 baseline.